SIMATS ENGINEERING
ASSESSMENT TOOL 3

Course Code	CSA0702	**Course Title**	Computer Networks
CO Assessed	CO2	**Assessment**	Debugging & Optimising Task
Weightage		**Total Marks**	

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Section/Batch	AIDS-2023	**Date of Submission**	
Faculty Incharge	DR.R S Selvan	**Assessment**	Individual Submission

TOTAL MARKS	**20**	**OBTAINED MARKS**	

17/20

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 30%

Code of Conduct:

I V.Sree Ruthin Reddy (192324112) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

V. Re

Annexure A

Debugging & Optimisation Task

1. A network administrator has configured a workstation with the following settings:

- **IP Address:** 192.168.10.65
- **Subnet Mask:** 255.255.255.192 (/26)
- **Default Gateway:** 192.168.10.1

However, the workstation is unable to communicate with other systems configured in the subnet **192.168.10.0/26**.

Task

Debug the network configuration by:

- Identifying the subnet to which the configured IP address belongs.
- Determining the network address, broadcast address, and valid host range.
- Verifying whether the configured IP address and gateway are valid.
- Identifying the configuration errors.
- Optimizing the configuration by assigning an appropriate IP address and default gateway.
- Calculating the total number of usable host addresses after correction.

2. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

3. An organization has allocated the network **192.168.1.0/24** into several departments using the subnet masks **/26**, **/27**, and **/28**. Some departments report insufficient IP addresses, while others have a large number of unused addresses.

Task

Analyze and optimize the subnet allocation by:

- Identifying the network address, broadcast address, and host range for each subnet.
- Calculating the number of usable and unused IP addresses.
- Identifying inefficient subnet allocations.
- Debugging the addressing plan by locating subnet allocation mistakes.
- Recommending an optimized subnetting scheme using VLSM to improve address utilization while supporting current departmental requirements.

4. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

5. A router contains the following routing table.

Destination Network	**Next Hop**
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route	10.0.0.1

Users report that packets destined for **192.168.2.100** are not reaching the intended network.

Task

Debug the routing configuration by:

- Identifying the matching routing entry using the longest-prefix match.
- Determining the next-hop router selected for packet forwarding.

Problem Identification.

1. Configured IP address 192.168.10.65 belongs to Subnet 192.168.10.64/26.
2. default gateway 192.168.10.1 belongs to subnet 192.168.10.0/26.
3. Since workstation & gateway are in diff subnets.

Debugging :

Calculate subnet : Subnet mask = /26.

Block size = 64

Subnet size = 64

Subnet	Address Range
192.168.10.0/26	0-63
192.168.10.64/26	64-127.
192.168.10.128/26	128-191
192.168.10.192/26	192-255.

Optimisation

Assign both the workstation & gateway to the same subnet.

Device	Correct address
Gateway	192.168.10.65.
Analysis	192.168.10.66.

1. communication failure occured because the gateway belonged to different subnet.

find optimal configuration.

Network address - 192.168.10.64

Broadcast address - 192.168.10.127.

Host range - 192.168.10.65 - 192.168.10.126.

Gateway - 192.168.10.65

Workstation - 192.168.10.66.

2. Problem Identification

Given

Device A: 2001:db8:ff00:42:8329/64

Device B: 2001:db8:00:1::1:/64

Debugging:

expanded express

Device A:

2001:0db8:0000:0000:ff00:0042:8329.

Device B:

2001:0db8:0000:000:0001:0000:0000:0001

Network prefix

with /64

Device A

001:0db8:0000:0000:0001:0000:0000:0001.

Device B

2001:0db8:0000:0001::/64

Both belong to same subnet

Problem:

No prefix mismatch.

Communication failure may due to

1. Firewall
2. Router configuration

Optimization

Example address

Device A:

2001: db8 : 10/64

Device B:

2001: odb8 : 0000 : 0000 :: /64

Belong belong to same Subnet

Analysis:

Both devices already share the same network prefix. IPv6 addressing is correct. network trouble shooting should focus.

3. Probeem Identification

Given:

192.168.1.0/24

Departments
1. 126
2. 127
3. 128

Some department have excess address.

Debugging

1. 126

Network - 192.168.1.0.

Host range - 192.168.1.1 - 192.168.1.62

Broad cast - 192.168.163

usable Hosts - 62

2. 127

network - 192.168.1.64 Host range - 192.168.165 - 192.168.1.a U

Broadcast - 192.168.1.95

usable hosts - 30.

128

network - 192.168.1.96.

Host range - 192.168.197 - 192.168.1.11

Broadcast - 192.168.1.11

usable Hosts - 14.

Analysis

Department need only 10 hosts.

allocated 126

Unused = 62-10

= 52 hosts wasted.

4. Expanded addresses.

Device A:

2001:0db8:0000:0000:0001:0000:0000:000)

Device B:

2001:0db8:0000:0000:0001:0000:0000:001

available Hosts

2^{64} = 18,446,744,073,709,551,616.

5. problem Identification

Defination	next Hop
192.168.1.0/24	1.0.0.02
192.168.2.0/24	1.0.0.3
default	1.0.0.0.1

Destination IP:

192.168.2.100

192.168.2.100 is valid.